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Chapter 6

Adding Data Fields



Why Read This?



Tip: This is a Tutorial. Read this to understand how to extend the list of Variables (Data Fields) available within Constructor.



Note: There are separate appendices called: Variable Substitution Skeleton Program and Data Field List.

What Are Data Fields?

Within the context of Work Management, Data Fields are values that are evaluated at run-time. They are used, for example, in messages and Emails. They are often called Variables.



Note: There is a set of Data Fields issued with the software on the CD.



Note: When exception or Email messages are defined within Constructor, they may include any of the available pre-defined Data Fields.

Four Types Of Data Fields



Warning: You can safely add Text and SQL types. However, take great care with Programmatic types and do not use Workflow types unless instructed by JBA.

Text

Literal text can be defined up to 256 characters long.

SQL

A Data Field is evaluated by executing an SQL statement. SQL statements may themselves contain variables hence resolving a Data Field may require recursive variable resolution. The SQL statement is defined in Constructor.

Programmatic



Warning: If you add a Programmatic type of Data Field then you need to create an RPG program with a subroutine for each Data Field. Please refer to the appendices called Variable Substitution Skeleton Program and Data Field List to help you with this.

Calls program defined in Constructor to evaluate Data Fields. All programs written to evaluate run-time Data Fields will be of the same structure as the Skeleton program QRPGRSRC/@SWMVARSUB. Output, P#VARV.

Note that one program may be used to evaluate many Data Fields. Each Data Field will have its own subroutine within its program.

Workflow



Warning: Workflow Data Fields need to be supported by the Work Management Engine. You will only need to amend these under exceptional circumstances and with guidance from JBA.

These are the Work Management information Data Fields:

- Company
- Business Process
- Business Object
- Document
- Activity
- Current Time
- Current Date.

How To Add More Data Fields

This explains the general process:

1. In Constructor select Data Fields from the Work Management Menu. This displays the Maintain Data Field Window. This lists all the Document Types.
2. You Select a Document Type and all the Data Fields for that Document Type are listed.
3. Note that you can maintain this list by using the New, Edit and Delete buttons at the bottom of the window. Selecting the New button displays the New Data Field Window. Selecting the Edit button displays the Edit Data Field Window. These two windows have the same fields but just have different titles.
4. Select the type of Data Fields: Text, SQL, Programmatic and Workflow.

There are three tutorials here showing how to add different types of Data Fields in Constructor.



Warning: If you add a Programmatic type of Data Field then you need to create an RPG program with a subroutine for each Data Field. Please refer to the appendices called Variable Substitution Skeleton Program and Data Field List to help you with this.

Add A Programmatic Data Field

Problem

In this example the Customer Credit Usage field is needed to create an Action Agent Path and does not appear in the browse list for the Input Condition field. As Customer Credit Usage is not a standard field on the System 21 database, an RPG program has been written called AA1001 to calculate the value.

This is how you should see it:

Code	Combination	Input Condition	Operator	Test Condition	Display Name
2	1st Rule	Customer Credit Usa...	>	100	Over Limit
(Default)					Default

This is what to do to create the Data Field:

1. Select Data Fields from the Work Management Menu. This displays the Maintain Data Fields Window.

Document type	Data field name	Description	Field type	Usage	Data field
PICKNOTE	AcknowledgePrtFlag	Acknowledge Print Flag	S	D	Select PI
SALESDORDER	Activity	Activity	W	D	
	Agent	Agent	S	D	Select AI
	BaseOrderValue	Base Order Value	S	D	Select BI
	BusinessObject	BusinessObject	W	D	
	BusinessProcess	BusinessProcess	W	D	
	Carrier	Carrier	S	D	Select C
	Company	Company	W	D	
	CurrencyCode	Currency Code	S	D	Select CI
	CurrencyRate	Currency Rate	S	D	Select CI
	CurrencyRateCode	Currency Rate Code	S	D	Select CI
	CurrentDate	CurrentDate	W	D	
	CurrentTime	CurrentTime	W	D	
	CustAccomInvoiceFlag	Customer Accompanying Invoice F...	S	D	Select AI
	CustAcknowledgement	Customer Acknowledgment	S	D	Select AI
	CustAcMgr	Customer Account Manager	S	UD	Select PI
	CustAddressLine1	Customer Address Line 1	S	D	Select C
	CustAddressLine2	Customer Address Line 2	S	D	Select C
	CustAddressLine3	Customer Address Line 3	S	D	Select C
	CustAddressLine4	Customer Address Line 4	S	D	Select C
	CustAddressLine5	Customer Address Line 5	S	D	Select C
	CustAgent	Customer Agent	S	D	Select AI
	CustAllowDaysOutTerm	Customer Allowed Days Out Term	S	D	Select CI

New... Edit... Delete Close

2. Select New to display the New Data Field Window.
3. Enter the details for CUSTOMERCREDITUSAGE field.

Data field name: CUSTOMERCREDITUSAGE Document type: SALESDORDER Data field type: Programmatic

Description: Customer Credit Usage

Program: AA1001

Usage:

- ☐ Role
- ☐ User
- ☐ E-mail address
- ☐ Priority
- ☒ Description

Data type: Packed decimal numeric Field length: 5 Decimal places: 2

OK Cancel

4. Click OK and the details appear in the list for SALESDORDER Document Type.
- Note that the CUSTOMERCREDITUSAGE field has a Usage set to Description to restrict when this variable will be allowable. The Usage Types are:
- Role - variable can be used when selecting the Role recipient type of schedule link.
 - User - variable can be used when selecting the User recipient type of schedule link.
 - Email - variable can be used when selecting the Email recipient type of schedule link.

Priority - variable can be used when selecting the Priority of a schedule link.

Description - variable can be used in general message areas and title overrides.

The Data Field name is the name of the variable that will be used on the Server and is limited to 20 alpha characters.

The Document Type Field will restrict when this definition of this variable will be available. Thus a Manual Activity with a Document Type of SALESORDER will only display variables with a matching Document Type. This is because the definition of how you retrieve a variable is driven by the key data available to you - if processing a Sales Order, the SL statement to retrieve the Customer Name will be different to the SQL statement if processing a Purchase Order.

The Description Field is a 35 alpha field holding a long description of the variable. This is the name that will be used throughout Constructor.

Document Type	Data Field Name	Description	Field Type	Usage	Data Field
PICKNOTE	CustSettleProfCode	CustSettleProfCode	S	D	Select S
SALESORDER	CustTelexNumber	CustTelexNumber	S	D	Select T
SALESORDER	CustBankDetails	CustBankDetails	S	D	Select E
	CustPaymentMethod	CustPaymentMethod	S	D	Select F
	CustCountryCode	CustCountryCode	S	D	Select C
	CustPreferCreditLmt	CustPreferCreditLmt	S	D	Select F
	CustAlternativeName	CustAlternativeName	S	D	Select A
	CustPrimeCurrRateCde	CustPrimeCurrRateCde	S	D	Select C
	CustAlternativePhone	CustAlternativePhone	S	D	Select F
	CustAlternativeFax	CustAlternativeFax	S	D	Select F
	CustAlternativeCntct	CustAlternativeCntct	S	D	Select C
	CustStmntBalance	CustStmntBalance	S	D	Select E
	CustLastStmntRunNo	CustLastStmntRunNo	S	D	Select L
	CustLastStmntDate	CustLastStmntDate	S	D	Select C
	CustCurrentBalance	CustCurrentBalance	S	D	Select E
	CustOrdersQValue	CustOrdersQValue	S	D	Select E
	CustDebtorDays	CustDebtorDays	S	D	Select C
	CustAverageDaysTaken	CustAverageDaysTaken	S	D	Select A
	CustCurrentDebtorDys	CustCurrentDebtorDys	S	D	Select C
	EmailAccounts	EmailAccounts	S	D	Select F
	EmailCSM	EmailCSM	S	D	Select F
	CustAcMgr	CustAcMgr	S	U	Select F
	CUSTOMER CREDIT USAGE	Customer Credit Usage	W	D	Select F

New Edit Delete Cancel

5. The details have already been saved so you can select Cancel to close the Window.

Add An SQL Data Field

Problem

In this example the scheduling details are being added for the Account Manager Contact Customer activity. This involves passing an activity on to another user so you set up a User Schedule. The activity will then be passed on to the Account Manger and will appear in the Action List for any System 21 Explorer user with the Role of Account Manager.

In this example the CustAcMgr field is needed for the Select Schedule Window in the Recipient field. This is how it should look below:

The screenshot shows the 'Execution Details' dialog box with the following settings:

- Schedule Type:**
 - ☒ User Controlled
 - ☐ Automatic Batch
 - ☐ Other
 - ☐ Automatic Immediate
 - ☐ Scheduled Batch
- ☐ Multiple object processing
- Tabbed Interface:**
 - Schedule Tab:**
 - Recipient Type: User
 - Document Type: SALESORDER
 - Recipient: Customer Account Manager
 - Static: ☐ DataField: ☒
 - Roles Tab:** (Empty)
 - Action List Tab:**
 - Priority: Normal
 - Static: ☒ DataField: ☐
- Buttons:** Ok, Cancel

If it doesn't appear in the Recipient list then this is what to do:

1. Select Data Fields from the Work Management Menu. This displays the Maintain Data Fields Window.

Document type	Data field name	Description	Field type	Usage	Data field
PICKNOTE	AcknowledgePrtFlag	Acknowledge Print Flag	S	D	Select PI
SALESDORDER	Activity	Activity	W	D	
	Agent	Agent	S	D	Select AI
	BaseOrderValue	Base Order Value	S	D	Select BI
	BusinessObject	BusinessObject	W	D	
	BusinessProcess	BusinessProcess	W	D	
	Carrier	Carrier	S	D	Select C
	Company	Company	W	D	
	CurrencyCode	Currency Code	S	D	Select CI
	CurrencyRate	Currency Rate	S	D	Select CI
	CurrencyRateCode	Currency Rate Code	S	D	Select CI
	CurrentDate	CurrentDate	W	D	
	CurrentTime	CurrentTime	W	D	
	CustAccomInvoiceFlag	Customer Accompanying Invoice F...	S	D	Select AI
	CustAcknowledgement	Customer Acknowledgment	S	D	Select AI
	CustAcMgr	Customer Account Manager	S	UD	Select PI
	CustAddressLine1	Customer Address Line 1	S	D	Select C
	CustAddressLine2	Customer Address Line 2	S	D	Select C
	CustAddressLine3	Customer Address Line 3	S	D	Select C
	CustAddressLine4	Customer Address Line 4	S	D	Select C
	CustAddressLine5	Customer Address Line 5	S	D	Select C
	CustAgent	Customer Agent	S	D	Select AI
	CustAllowDaysOutTerm	Customer Allowed Days Out Term	S	D	Select CI

New... Edit... Delete Close

2. Select New to display the New Data Field Window.
3. Enter the details for CustAcMgr field.

Data field name: CustAcMgr Document type: SALESDORDER Data field type: SQL

Description: Customer Account Manager

SQL statement:
Select PRMD15 from osld1f3/INP15, osld1f3/DEP40 where CONO40 =
CONO15 and PRMT15 = 'SLMN' and PSAR15 = CUSN40 and CONO40 =
Company and ORDN40 = Document

Usage:
☐ Role
☒ User
☐ E-mail address
☐ Priority
☒ Description

Data type: Alpha string Field length: 30 Decimal places: 0

OK Cancel

4. Click OK and the details appear in the list for SALESDORDER Document Type. Note that the CustAcMgr field has a Usage for User because it should only appear as a data field when prompting for users.

Document type	Description	Field type	Usage	Data field information
PICKNOTE	Customer Account Manager	S	UD	Select PRMD15 from osld1f3/INP...
SALESORDER	Admin E-mail	T	E	z1admin@z1.co.uk
	Acknowledge Print Flag	S	D	Select PRNA40 from OEP40 wher..
	Activity	W	D	
	Agent	S	D	Select AGNT40 from OEP40 wher..
	Base Order Value	S	D	Select BORV40 from OEP40 wher..
	BusinessObject	W	D	
	BusinessProcess	W	D	
	Carrier	S	D	Select CARR40 from OEP40 wher..
	Company	W	D	
	Currency Code	S	D	Select CURN40 from OEP40 wher..
	Currency Rate	S	D	Select CURT40 from OEP40 wher..
	Currency Rate Code	S	D	Select CURC40 from OEP40 wher..
	CurrentDate	W	D	
	CurrentTime	W	D	
	Customer Accompanying Invoice Flag	S	D	Select AINF20 from OEP40,OEP2...
	Customer Acknowledgment	S	D	Select ACKN20 from OEP40,OEP2...
	Customer Address Line 1	S	D	Select CAD105 from OEP40,SLP0...
	Customer Address Line 2	S	D	Select CAD205 from OEP40,SLP0...
	Customer Address Line 3	S	D	Select CAD305 from OEP40,SLP0...
	Customer Address Line 4	S	D	Select CAD405 from OEP40,SLP0...
	Customer Address Line 5	S	D	Select CAD505 from OEP40,SLP0...
	Customer Agent	S	D	Select AGNT20 from OEP40,OEP...

Buttons: Add... Edit... Delete Close

5. The details have already been saved so you can select Cancel to close the Window.

Add A Text Data Field

Problem

In this example scheduling details are being added for the Advise Customer Near Limit activity.

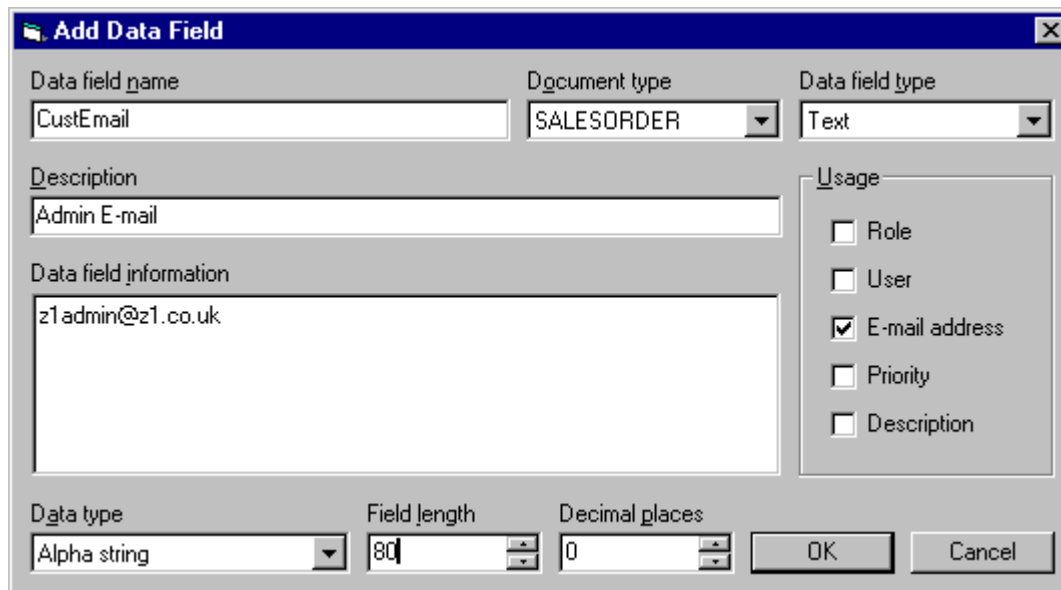
This will send a message by Email to the customer to advise them that they are near their credit limit. First you need to add a data field called CustEmail for the Email address.

1. Select Data Fields from the Work Management Menu. This displays the Maintain Data Field Window.

Document type	Data field name	Description	Field type	Usage	Data field
PICKNOTE	%CreditLimitUsed	%CreditLimitUsed	P	D	AA1001
SAELSDORDER	AcknowledgePrtFlag	AcknowledgePrtFlag	S	D	Select PI
SALESORDER	Activity	Activity	W	D	
	Agent	Agent	S	D	Select AI
	BaseOrderValue	BaseOrderValue	S	D	Select BI
	BusinessObject	BusinessObject	W	D	
	BusinessProcess	BusinessProcess	W	D	
	Carrier	Carrier	S	D	Select C
	Company	Company	W	D	
	CurrencyCode	CurrencyCode	S	D	Select CI
	CurrencyRate	CurrencyRate	S	D	Select CI
	CurrencyRateCode	CurrencyRateCode	S	D	Select CI
	CurrentDate	CurrentDate	W	D	
	CurrentTime	CurrentTime	W	D	
	CustAccomInvoiceFlag	CustAccomInvoiceFlag	S	D	Select AI
	CustAcknowledgement	CustAcknowledgement	S	D	Select AI
	CustAcMgr	CustAcMgr	S	U	Select PI
	CustAddressLine1	CustAddressLine1	S	D	Select C
	CustAddressLine2	CustAddressLine2	S	D	Select C
	CustAddressLine3	CustAddressLine3	S	D	Select C
	CustAddressLine4	CustAddressLine4	S	D	Select C
	CustAddressLine5	CustAddressLine5	S	D	Select C
	CustAgent	CustAgent	S	D	Select AI

New Edit Delete Close

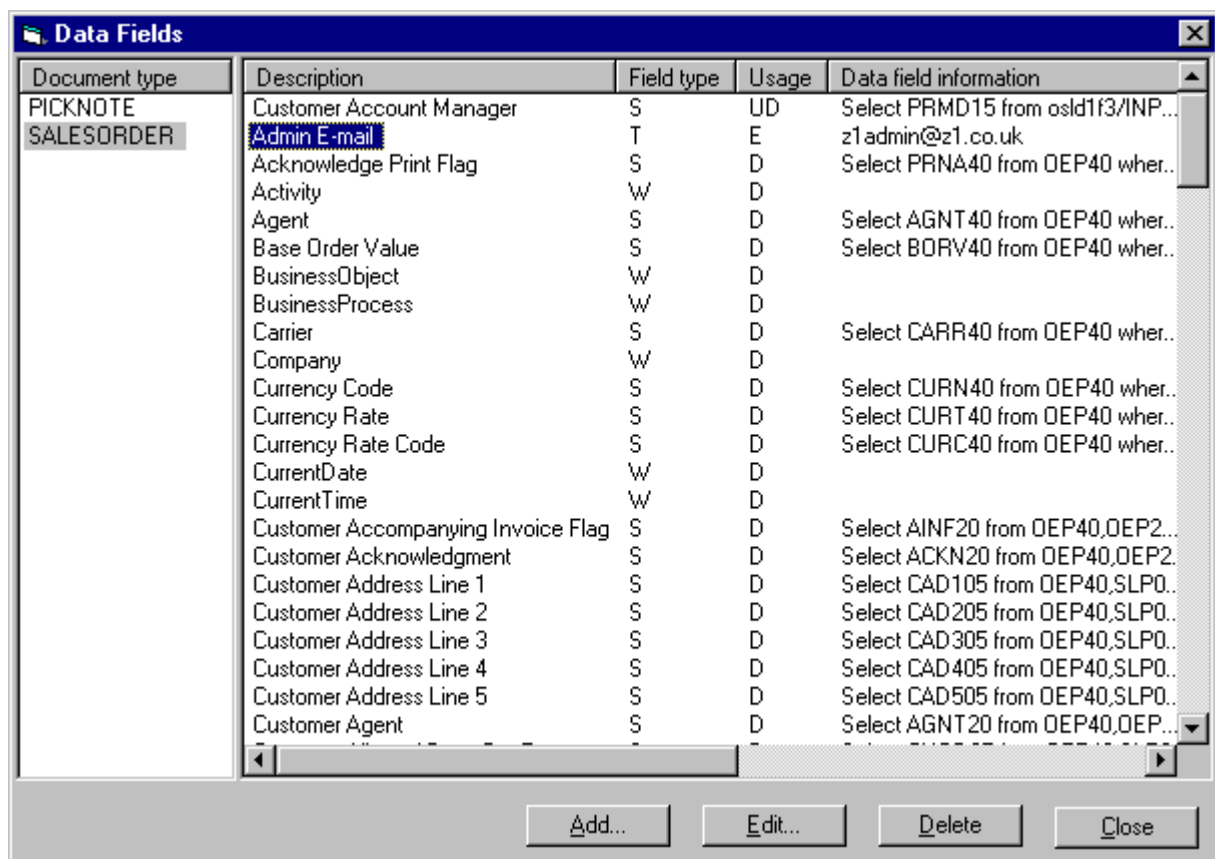
2. Select the New button. This displays the New Data Field Window
3. Enter the details as shown below:



The 'Add Data Field' dialog box is shown. It has the following fields and settings:

- Data field name:** CustEmail
- Document type:** SALESORDER
- Data field type:** Text
- Description:** Admin E-mail
- Data field information:** z1admin@z1.co.uk
- Usage:**
 - ☐ Role
 - ☐ User
 - ☒ E-mail address
 - ☐ Priority
 - ☐ Description
- Data type:** Alpha string
- Field length:** 80
- Decimal places:** 0
- Buttons:** OK, Cancel

4. Select OK to save the details and return to the Maintain Data Field Window.
5. Look for CustEmail. Note that this field has a Usage set to Email.



The 'Data Fields' window displays a list of data fields. The 'SALESORDER' document type is selected. The 'Admin E-mail' field is highlighted, showing its details in the right pane.

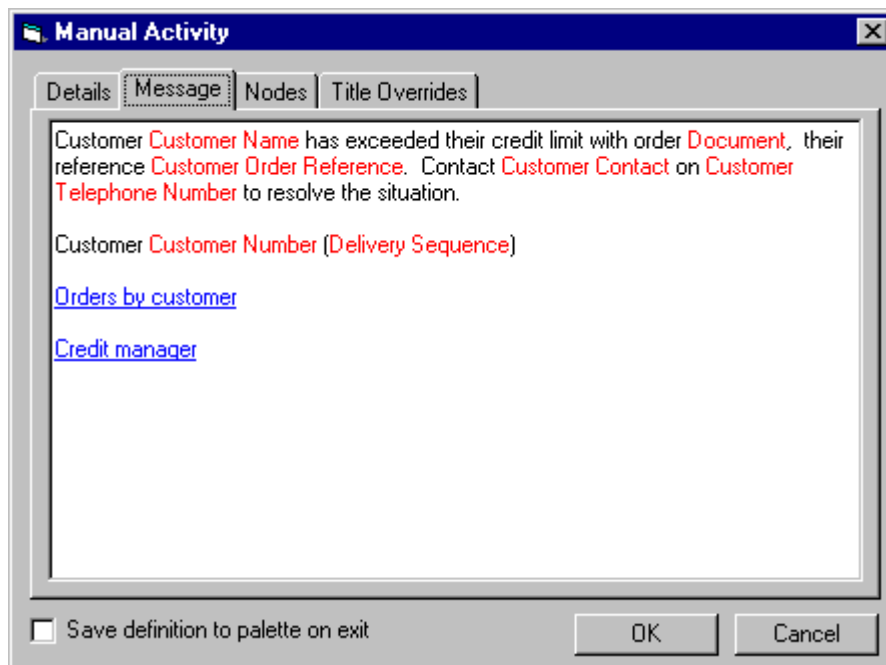
Document type	Description	Field type	Usage	Data field information
PICKNOTE	Customer Account Manager	S	UD	Select PRMD15 from osld1f3/INP...
SALESORDER	Admin E-mail	T	E	z1admin@z1.co.uk
	Acknowledge Print Flag	S	D	Select PRNA40 from OEP40 wher..
	Activity	W	D	
	Agent	S	D	Select AGNT40 from OEP40 wher..
	Base Order Value	S	D	Select BORV40 from OEP40 wher..
	BusinessObject	W	D	
	BusinessProcess	W	D	
	Carrier	S	D	Select CARR40 from OEP40 wher..
	Company	W	D	
	Currency Code	S	D	Select CURN40 from OEP40 wher..
	Currency Rate	S	D	Select CURT40 from OEP40 wher..
	Currency Rate Code	S	D	Select CURC40 from OEP40 wher..
	CurrentDate	W	D	
	CurrentTime	W	D	
	Customer Accompanying Invoice Flag	S	D	Select AINF20 from OEP40,OEP2...
	Customer Acknowledgment	S	D	Select ACKN20 from OEP40,OEP2...
	Customer Address Line 1	S	D	Select CAD105 from OEP40,SLP0...
	Customer Address Line 2	S	D	Select CAD205 from OEP40,SLP0...
	Customer Address Line 3	S	D	Select CAD305 from OEP40,SLP0...
	Customer Address Line 4	S	D	Select CAD405 from OEP40,SLP0...
	Customer Address Line 5	S	D	Select CAD505 from OEP40,SLP0...
	Customer Agent	S	D	Select AGNT20 from OEP40,OEP...

Buttons at the bottom: Add..., Edit..., Delete, Close

6. The details have already been saved so you can select Cancel to close the Window.

Using Data Fields In Messages

When you create a Manual Activity in Constructor you create a message for the user.
You complete the Message panel details on the Manual Activity Window as shown below.



This is what the user will see on the screen. The black text is actual text which the user sees but the red text is Variable Data, for example Customer Name.

To insert red text Data Fields use the right click menu and select Insert Data Field. If a field name does not exist then select the Maintain Data Field option and add to the list.



Note: The blue underlined text represents a button on the screen which the user can see and use. You add buttons by selecting Insert Button on the right click menu. This displays the Select Button Window. These buttons represent pathways which go off at a tangent to the processing and do not interfere in any way with the logical flow through the Business Process.

The Message text also appears on the tooltips for the activities as shown below.

